Design and Optimization of Nearest Neighbor Realization of Quantum Circuits

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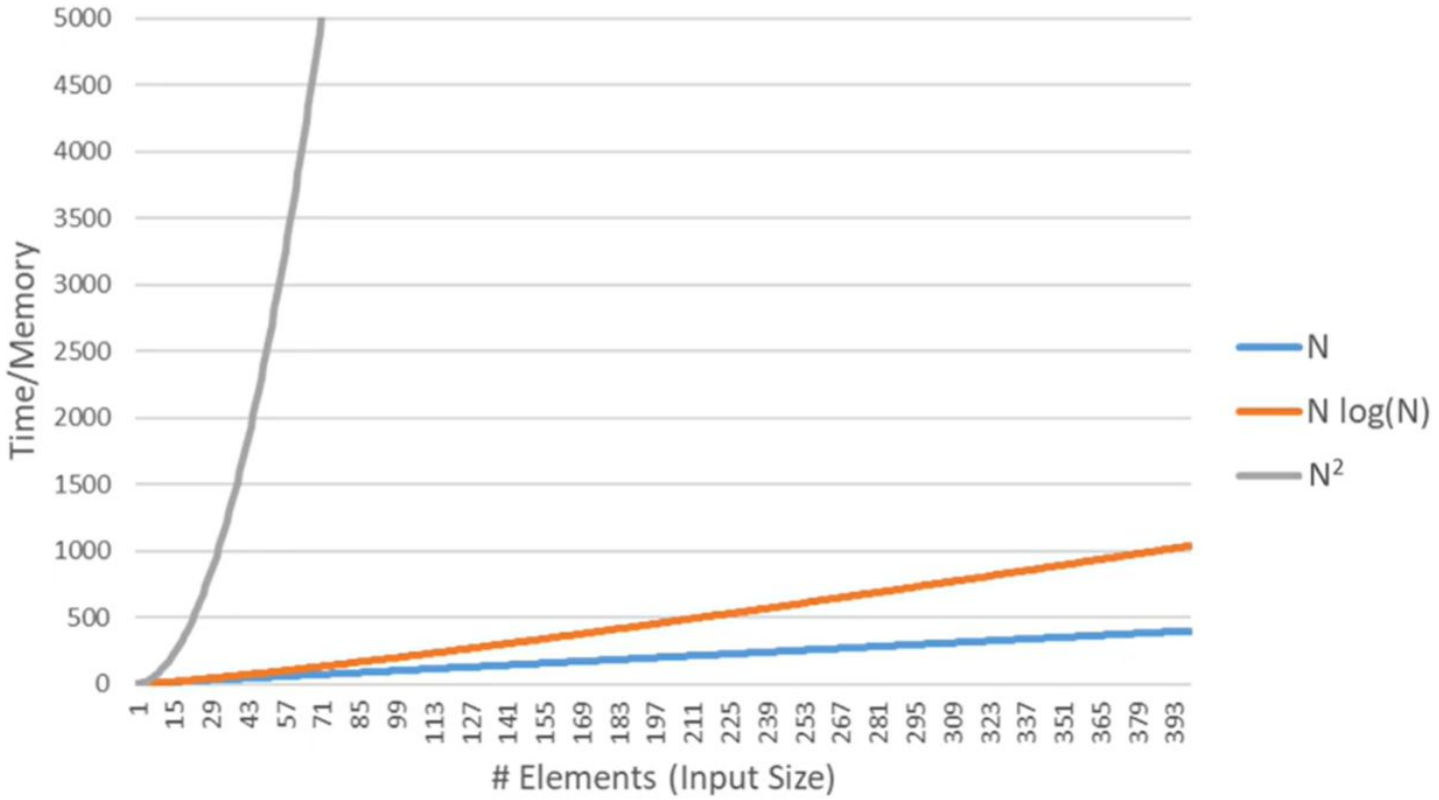
Outline of the Presentation

- **Introduction**
  - Quantum Computing
  - Terms & Terminologies
- **Nearest Neighbor (NN) Property**
  - Global Reordering
  - Local Reordering
  - Hybrid Reordering
- **Look-ahead Policy**
- **Nearest Neighbor Mapping**

Introduction

Motivation: there are intrinsically hard problems that today's computers cannot handle

Comparing Easy and Hard Problems



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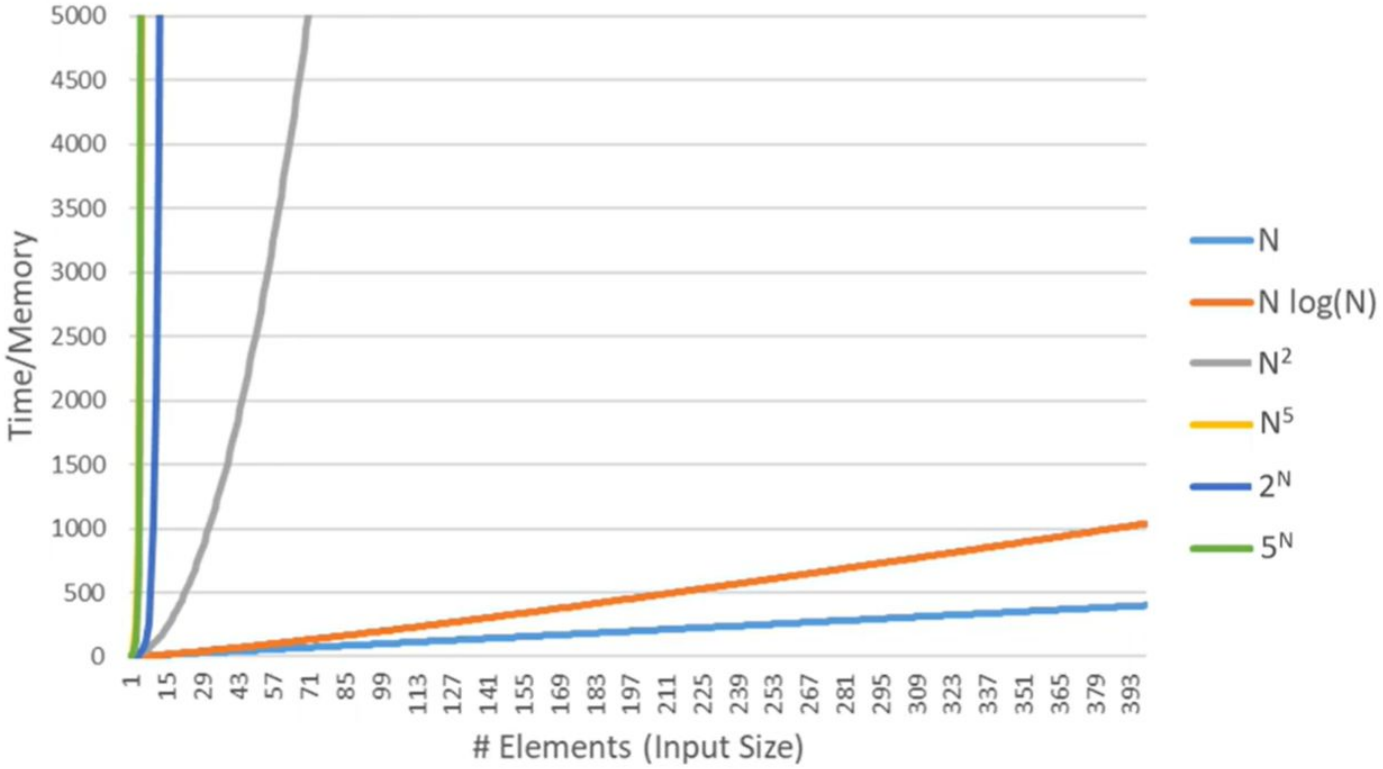

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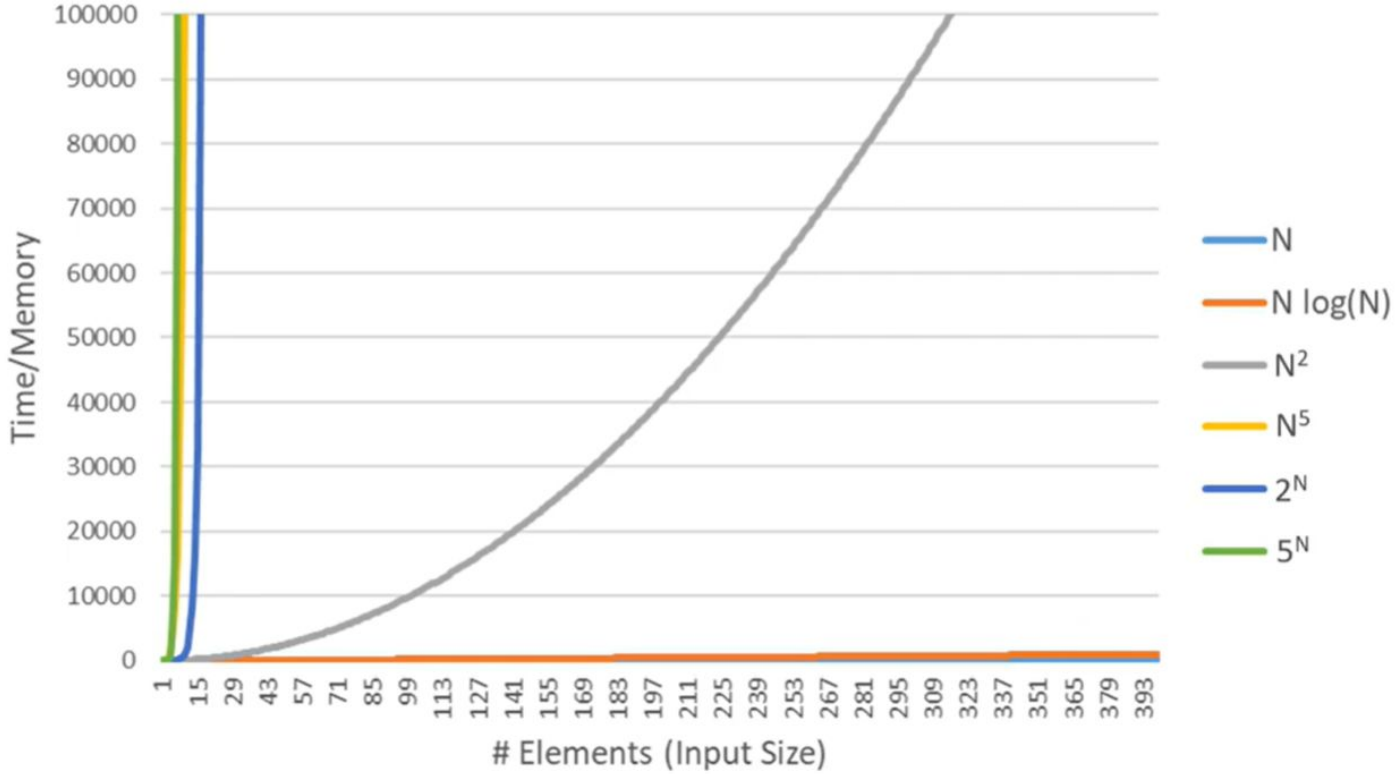

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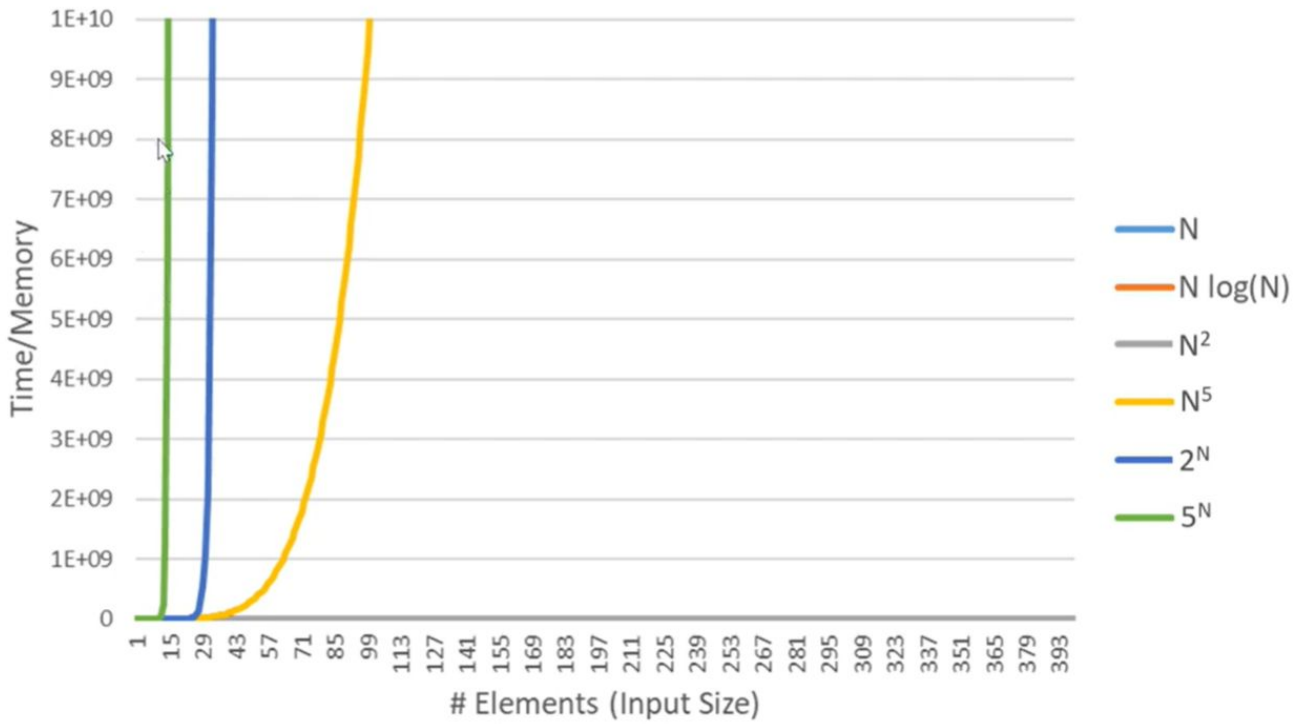

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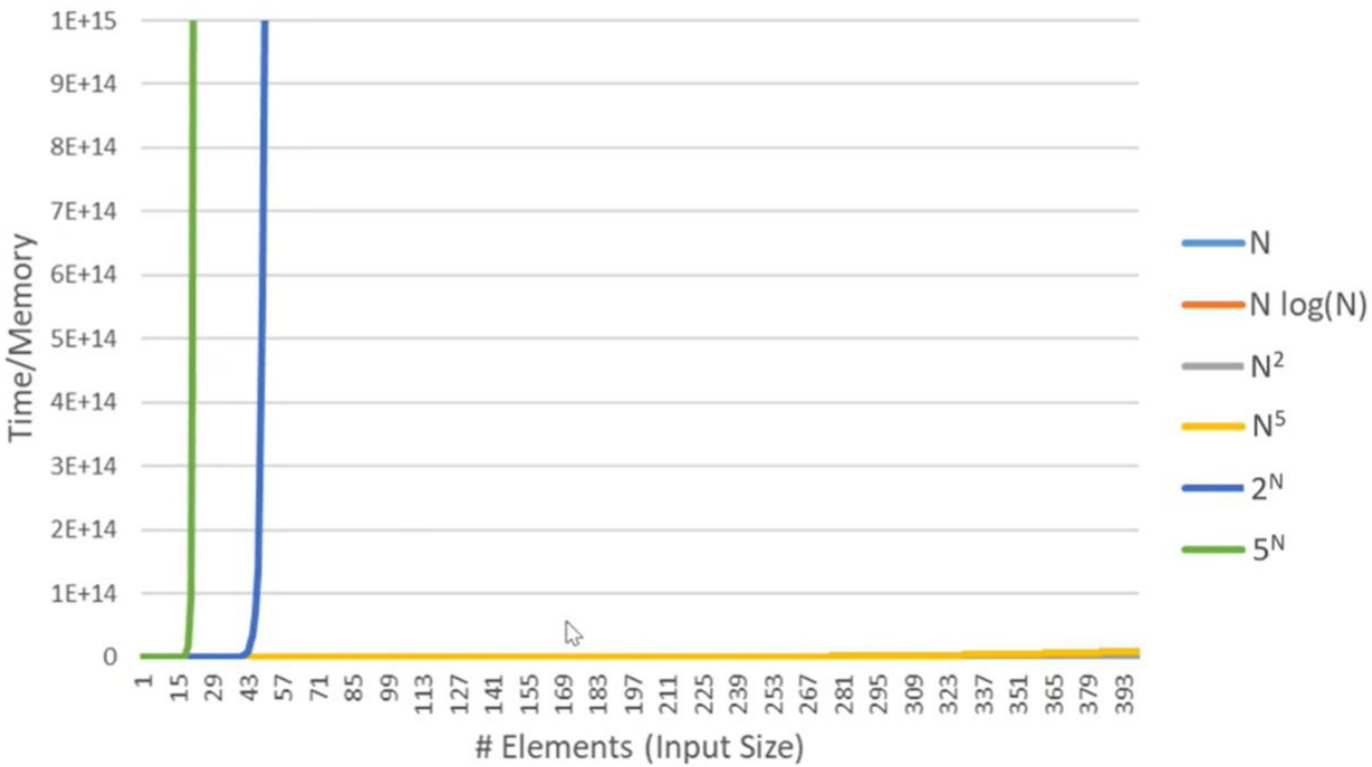

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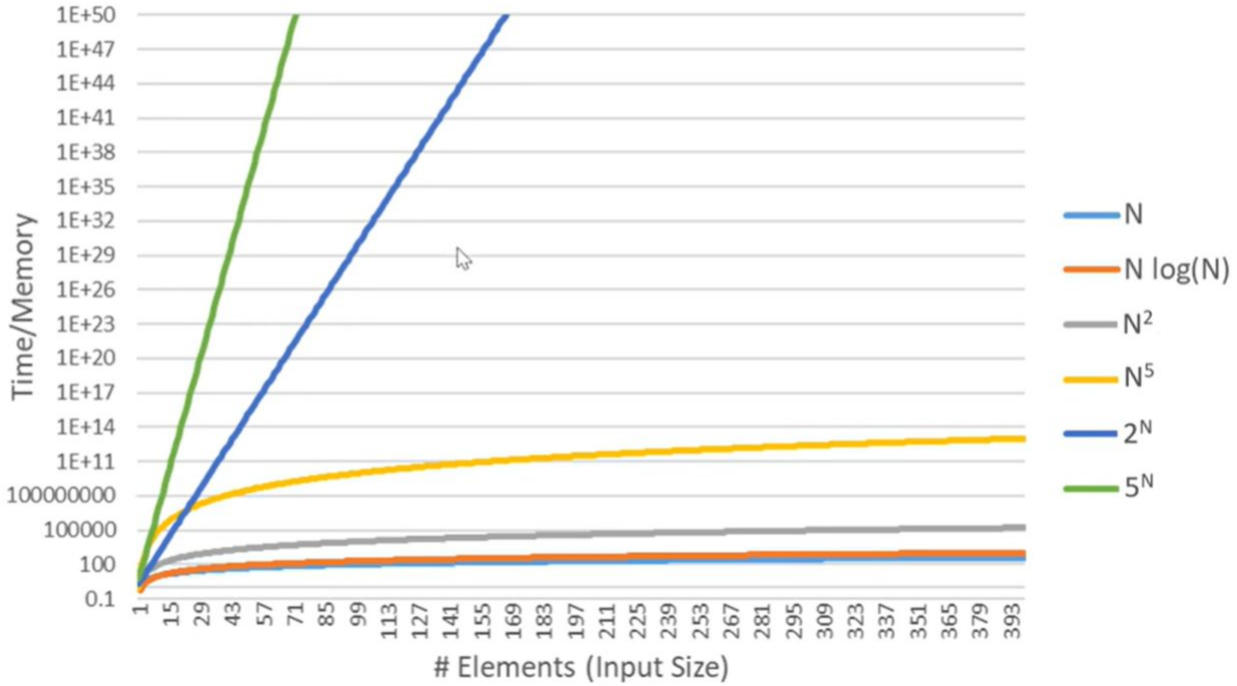
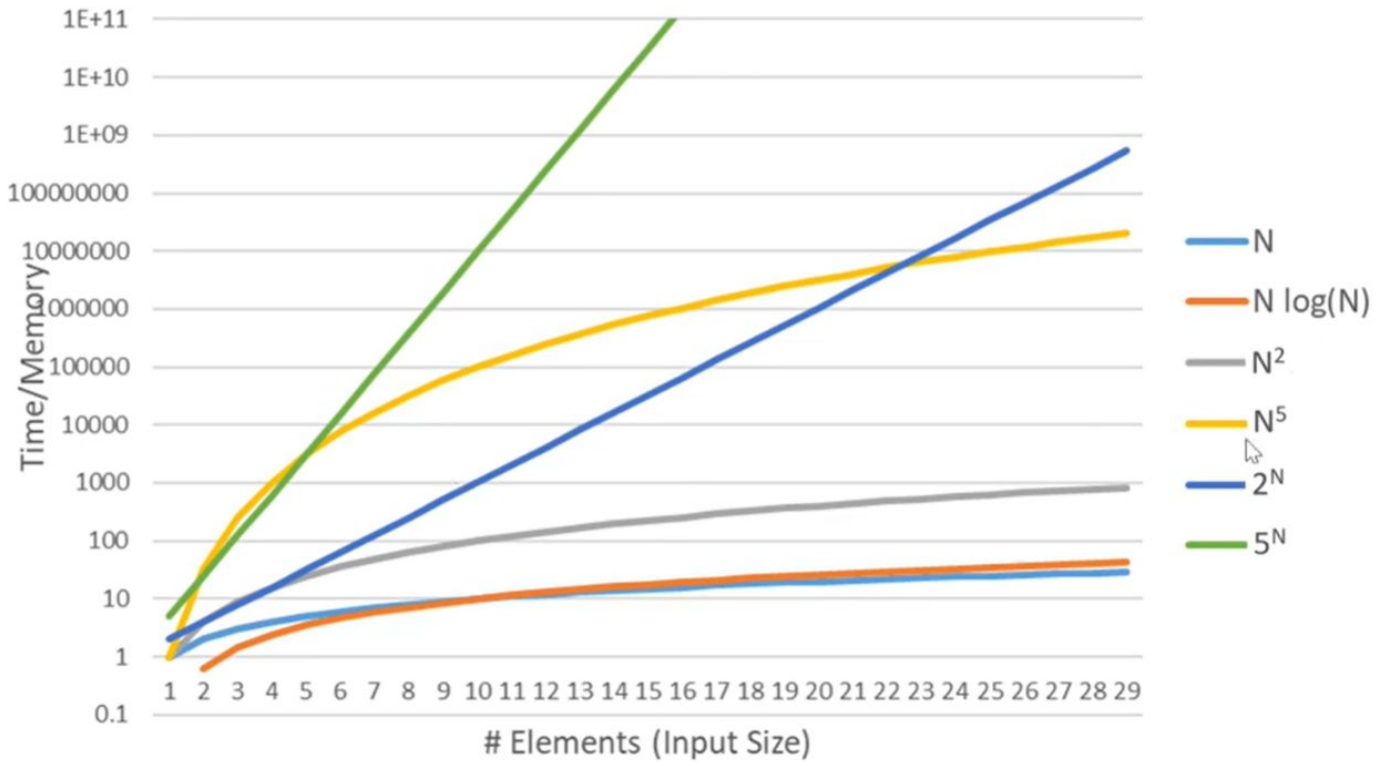
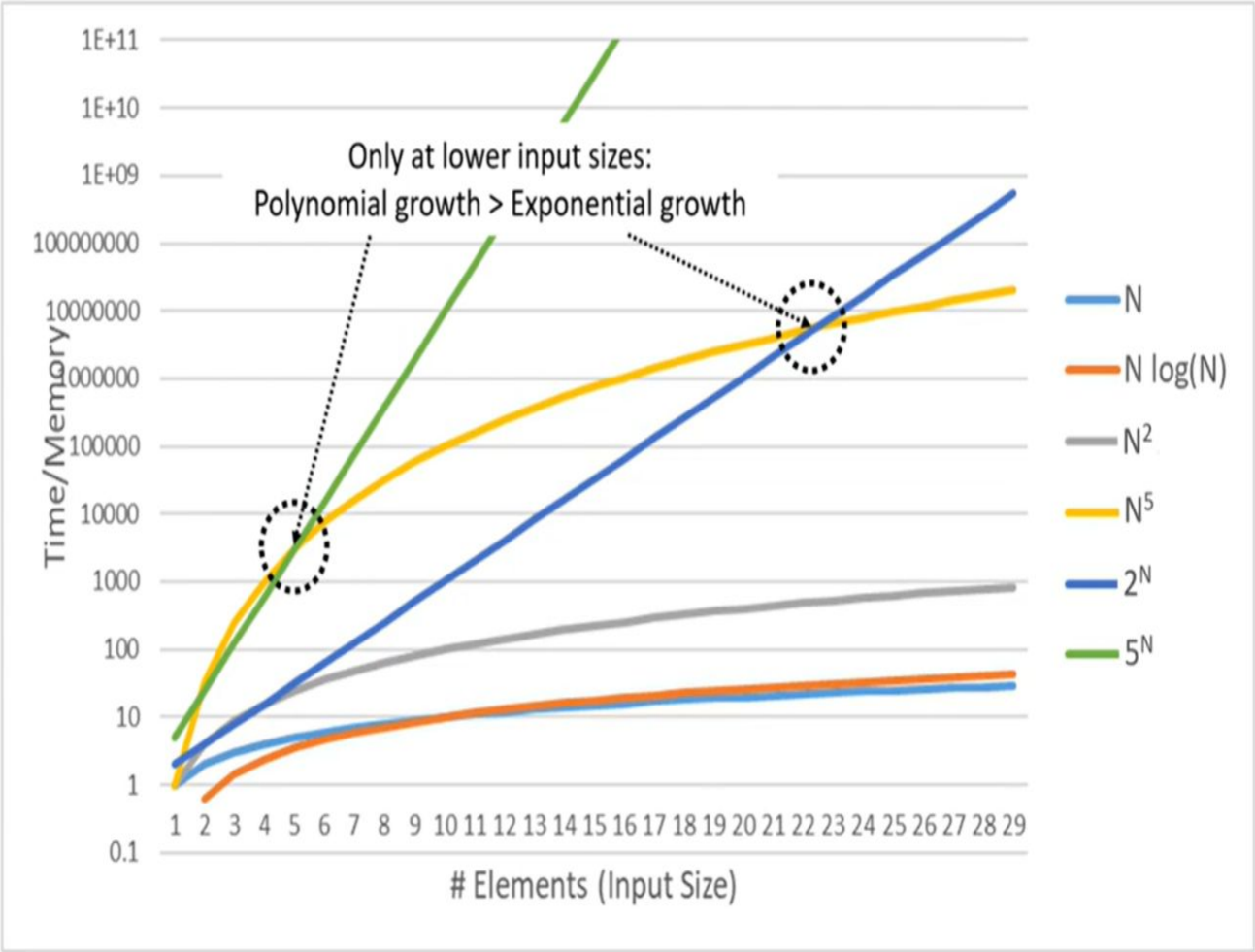


Comparing Easy and Hard Problems



Comparing Easy and Hard Problems



Issues in VLSI domain

- Power dissipation become a major problem in VLSI circuit.
- How to design information lossless circuit?

Solution to VLSI domain Problems

- Concept of Reversible Circuit evolves
  - Consumes small power
  - Ensures Information lossless in circuit realization
  - Desired requirement for quantum computation.

Quantum Computing

- Ability to solve intractable problems like discrete logarithms, prime factorization, database search more efficiently.
- Facilitates faster execution.

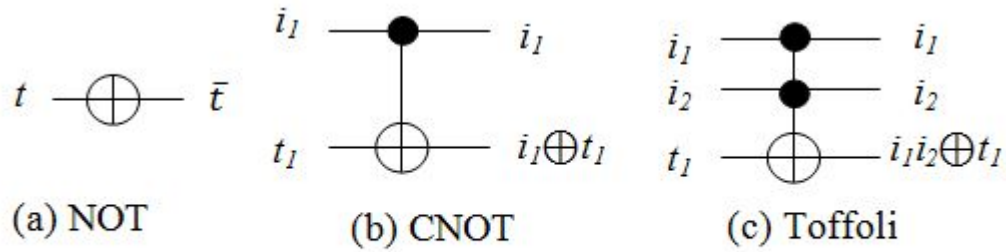
Terms and Terminologies

Reversible Circuit

- Digital circuit with a bijective mapping between the inputs and outputs and the number of inputs is equal to the number of outputs.
- A reversible logic circuit of size n with depth d is a cascade of d reversible gates over the set of input lines $X = \{ x_1, x_2, \ldots, x_n \}$, i.e. reversible circuit = $g_0 . g_1 . g_2 . \ldots . g_{d-1}$, where each g_i represent i^{th} reversible gate of the circuit.

Reversible Gate

- Basic components used to build reversible circuit following the reversible properties.



Basic reversible gates in Quantum computation

Qubit

Represents quantum information that can exist in the form as

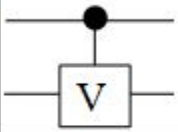
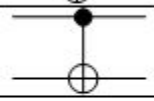
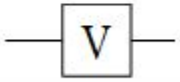
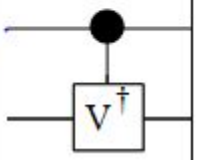
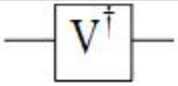
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \qquad (1)$$

where α and β represents the complex numbers and subjected to the constraint $\alpha^2 + \beta^2 = 1$

Quantum Gate

Building blocks of quantum circuit that performs manipulations on qubits to implement specific functions.

Table 1: Some basic quantum gates and their symbolic representations

Gates	Notation	Gates	Notation
NOT		Control led -V	
CNOT			
V		Control led -V†	
V†			

Quantum Cost

Number of quantum operations needed to realize a specific quantum functionality.

SWAP gate

Quantum gate that interchanges the states rather than values of the qubits.

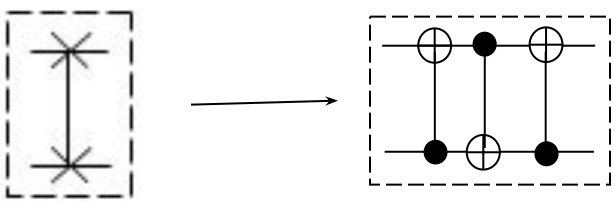
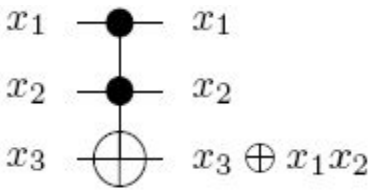
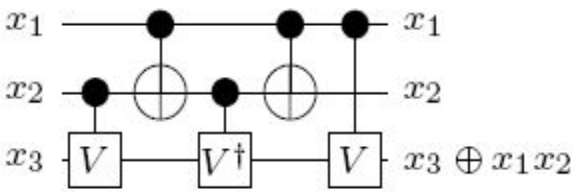


Fig. 1: SWAP gate representation

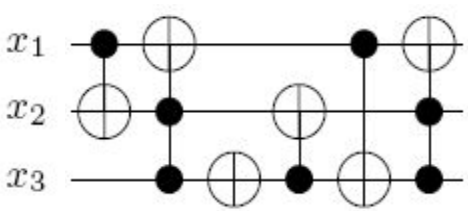
Reversible Circuit to Quantum Circuit

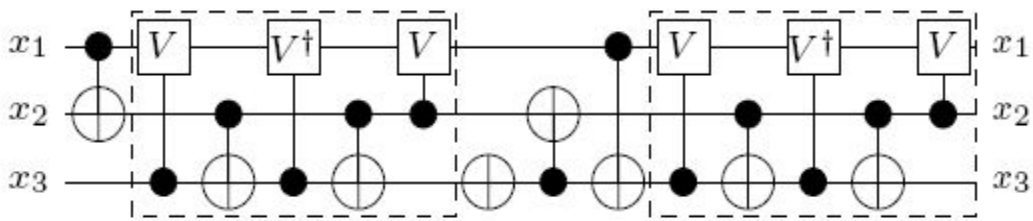


Reversible Toffoli gate



Technology Library





Technology library: *A library that is composed of elementary quantum gates. Like NCV which is composed of NOT, CNOT and V/V+ gate*

Nearest Neighbor (NN) Property

Nearest Neighbor Cost (NNC)

Interaction distance between the control (c) and target (t) qubits of any 2-qubit quantum gate (g) and can be represented mathematically as

$$NNC(g) = |c - t| - 1 \qquad (2)$$

SWAP gates are implemented to make any 2-qubit gate NNC free.

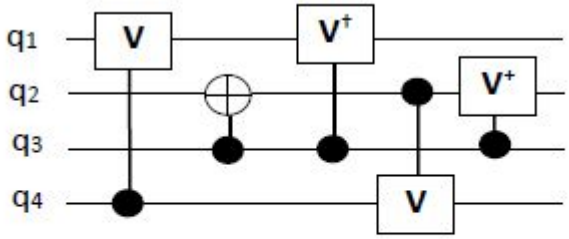


Fig. 2(a): Input circuit with NNC=4

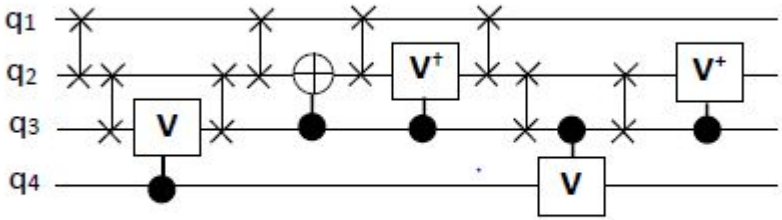


Fig. 2(b): NN design of the input circuit in Fig. 2(a)

Why NN design ?

- One of the major design constraints for physical implementation of quantum circuit is NN issue.
- Quantum circuit implementing technologies like ion-trap, superconducting qubits , demand the interacting qubits to be nearest neighbor one (NN value should be 0).

Nearest Neighbor Optimization

- SWAP gates are needed to design NN circuit.
- Causes cost overhead such as quantum cost and latency.
- Needs to implement SWAP gates efficiently.
- Two reordering schemes of **Global** and **Local** used for cost reduction.

Global Reordering

- Rearranges the original qubit positions and then applies SWAP gates before each non-adjacent gates
- Restores the previous qubit positions by again adding SWAP gates after each non-adjacent gates.

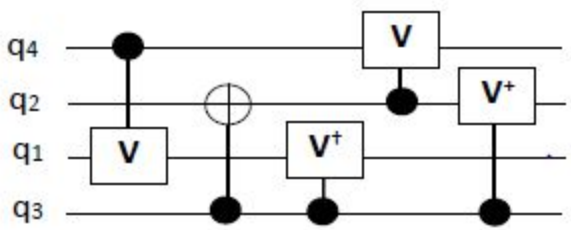


Fig. 3(a): Reordered circuit of Fig. 2(a) with NNC=3

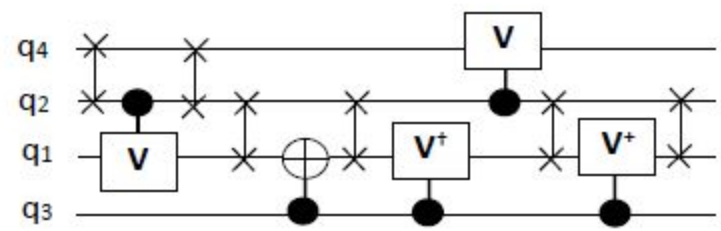


Fig. 3(b): NN-compliant circuit of Fig. 3(a) using global reordering

Local Reordering

- Discarded the condition of restoring the original qubit positions.
- Retains the change in qubit positions after SWAP gate implementations and applies it to the rest of the gates in the circuit.
- Reduces SWAP gates than the one used in global ordering.

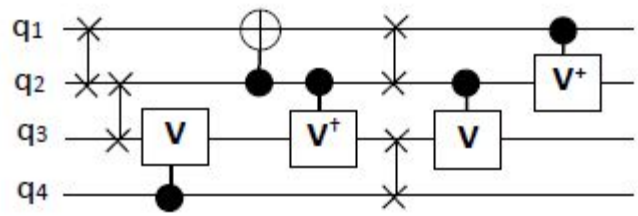


Fig. 4: NN design of Fig. 2(a) using local reordering

Hybrid Reordering

- Combination of Global and Local reordering.
- First Global reordering is applied on any arbitrary circuit then finally the resultant design solution is obtained by implementing Local reordering approach on Global solution.
- Requires lesser SWAP gates than other reordering approaches.

Look-ahead Policy

- Chooses the best possible SWAP insertion option.
- Determines the SWAP insertion based on the impact imposed on the following gates.
- To reduce the overhead involved in the approach, a circuit window is considered in which the evaluation is conducted over a specific number of gates.

Illustrative Example

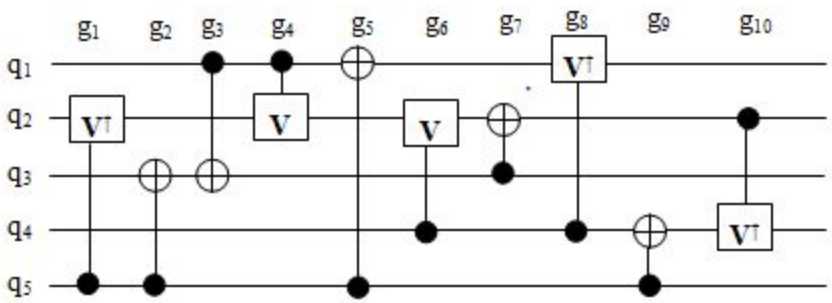


Fig. 5: Input circuit

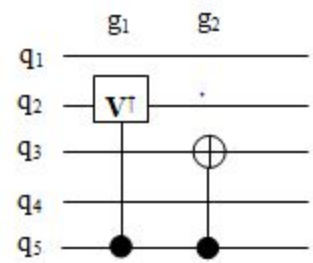


Fig. 6(a): Initial position of g_2

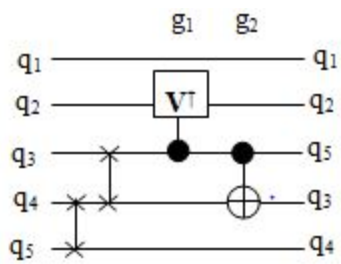


Fig. 6(b): Impact on g_2 (NNC=0) after S_1

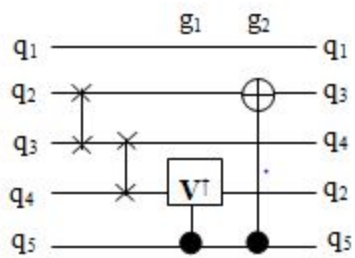


Fig. 6(c): Impact on g_2 (NNC=2) after S_2

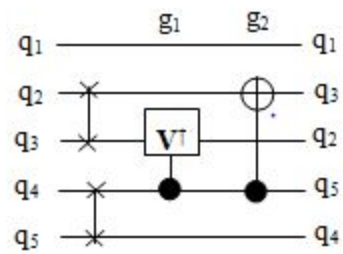


Fig. 6(d): Impact on g_2 (NNC=1) after S_3

Nearest Neighbor Mapping

- Can be implemented in various topological layouts- 1D, 2D and higher dimensions.
- Requires more SWAP count in 1D platforms than in other architectural layouts.
- 2D structures are more preferable than 3D as it challenges to manage high-density qubit interactions.

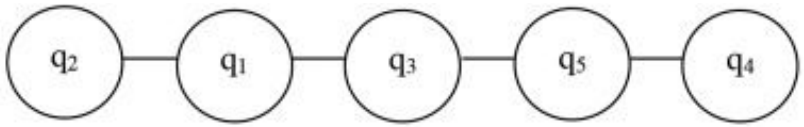


Fig. 7(a): Qubits arranged in 1D platform

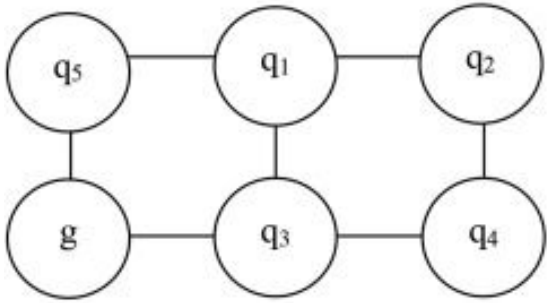


Fig. 7(b): Qubits arranged in 2D platform

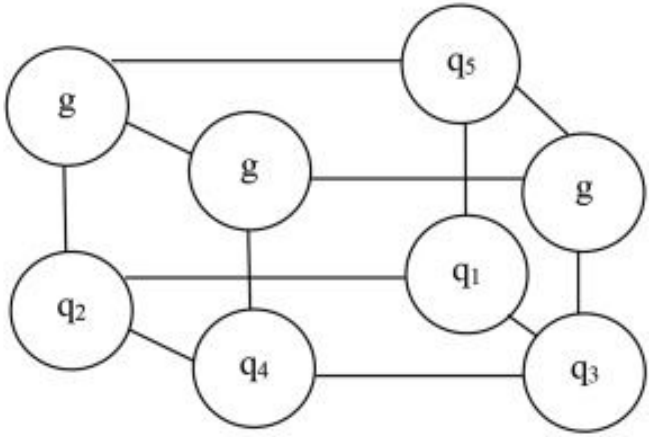


Fig. 7(c): Qubits arranged in 3D platform

Summary of the Session

- Discussed about Nearest Neighbor (NN) architecture
- Approaches followed to achieve the NN design.
- Different layout mapping of NN based quantum circuits.

Thank you